

Assessing the impact of wind generation on wholesale prices and generator dispatch in the Australian National Electricity Market



Sam Forrest ^{a,b,*}, Iain MacGill ^{c,d}

^a School of Economics, University of New South Wales, Sydney 2052, Australia

^b School of Mechanical and Manufacturing Engineering, University of New South Wales, Sydney 2052, Australia

^c School of Electrical Engineering and Telecommunications, University of New South Wales, Sydney 2052, Australia

^d Centre for Energy and Environmental Markets, University of New South Wales, Sydney 2052, Australia

H I G H L I G H T S

- Proposes methodologies to estimate short run impact of wind on electricity markets.
- Quantifies the merit order effect of wind generation on wholesale spot price.
- Wind is found to be significantly effecting gas fired generation.
- Evidence is found for wind having a notable impact on baseload coal generation.
- Discusses the implications for development of wind generation in Australia.

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Growing climate change and energy security concerns are driving major wind energy deployment in electricity industries around the world. Despite its many advantages, growing penetrations of this highly variable and somewhat unpredictable energy source pose new challenges for electricity industry operation. One issue receiving growing attention is the so-called ‘merit order effect’ of wind generation in wholesale electricity markets. Wind has very low operating costs and therefore tends to displace higher cost conventional generation from market dispatch, reducing both wholesale prices and conventional plant outputs. This paper extends the current literature on this effect through an empirical study employing a range of econometric techniques to quantify the impacts of growing wind penetrations in the Australian National Electricity Market (NEM). The results suggest that wind is having a marked impact on spot market prices and, while wind is primarily offsetting higher operating cost gas generation, it is now also significantly reducing dispatch of emissions intensive brown coal generation. Great care needs to be taken in extrapolating these results to longer-term implications, however, the study does propose a methodology for assessing this effect, highlights the impacts that wind is already having on NEM outcomes and suggests promising directions for future research.

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1. Introduction

Over the last decade, wind energy has emerged as one of the most successful and promising low-carbon electricity generation technologies worldwide. It is amongst the fastest growing sources of generation (REN21, 2011), has achieved significant penetrations in a number of countries (LBNL, 2011) and is widely agreed to be one of the key technologies for achieving a more sustainable future electricity sector (IEA, 2011). In general, the expansion of wind is being driven by its growing cost competitiveness, yet also by a varied mix of policies in support of energy security, environmental

and wider social outcomes. However, the growing deployment of this highly variable and somewhat unpredictable generation source poses new challenges for electricity industry operation. Any significant deployment of new generation within an electricity industry is likely to impact on the physical dispatch of existing plant, overall industry economics and commercial outcomes for participants. Wind energy is no exception and due to its unique operational and economic characteristics and the fact that it is the first intermittent generation to reach significant penetrations in large electricity grids, there is still much to learn about its technical, economic and commercial implications, and what type of electricity industry arrangements (and wider policy support frameworks) might best facilitate its deployment.

As well as the recent attention on the operational impacts of high wind penetrations (IEA Task 25, 2011), including balancing costs, reserve requirements, transmission and system capacity, there is a

* Corresponding author at: School of Economics, University of New South Wales, Sydney 2052, Australia.

Tel.: +614 31 901 296.

E-mail addresses: s.forrest@unsw.edu.au, sam.forrest@nera.com (S. Forrest).

growing body of literature exploring the potential impact of wind generation on commercial outcomes within wholesale spot markets. Two key issues that have emerged in this regard are the impact of wind output on spot price and the dispatch of other generation types.

The impact on price, or the ‘merit order effect’, results from low marginal cost wind output systematically displacing higher marginal cost generation through the least cost market dispatch process. It has been identified and its impact quantified in a range of markets around the world through descriptive (Cutler et al., 2011), simulation (Lamont, 2008; Sensfus et al., 2007) and empirical (Woo et al., 2011; Nicholson et al., 2010) studies.

The displacement of generation of various types by wind generation arises from this same merit order effect. The literature generally concludes that when wind is generating, dispatch of the higher operating cost, typically peaking, plant is the first to be impacted and that only under certain circumstances will baseload plant also be displaced (Green and Vasilakos, 2010b; Delarue et al., 2009; Goransson and Johnsson, 2009; Ummels, 2006).

To date, the focus has been on simulation studies with limited empirical work on the impact of wind generation being conducted. Although less common, empirical studies have particular importance because of the complexities in wholesale electricity markets that are difficult to capture in market simulations. While subject to their own set of assumptions and limitations, direct analysis of market data can provide valuable insights and is a vital supplement to other simulation work. In this paper we present an econometric analysis of the impact of wind generation on wholesale spot prices and outputs of other generation types within the South Eastern region of the Australian National Electricity Market (NEM). This region has a significant wind penetration by international standards (Cutler et al., 2011) and the wholesale market has high transparency with regard to historical prices and generating plant dispatch (MacGill, 2010).

We first develop an autoregressive linear model for the truncated half hourly spot price that can be used to estimate the merit order effect of the change in wholesale spot price per MWh of wind output. Second, we develop several simple linear models to assist in understanding the dynamic relationships that exist between wind output and baseload and peaking generation. The intent being to estimate how much of each generation type (primarily coal and gas plant in this case) can be considered to be displaced by wind output in the short run and the associated impact on greenhouse gas emissions from the electricity industry. The abatement issue is of particular relevance to South Eastern Australia where coal fired generation is from high emissions intensity brown coal (lignite) plants.

Section 2 of this paper outlines the theoretical underpinnings of our analysis and some key hypothesised outcomes for the impact of wind in wholesale electricity markets. It also describes some of the key simulation and empirical studies conducted to date. Section 3 provides background on the Australian NEM design, structure and policy context, and experience with wind industry integration. Section 4 outlines our model and the model development procedure for the price and dispatch analyses. Section 5 describes the data used in the analysis. Sections 6 and 7 outline the results of the price and dispatch analyses respectively and Section 8 provides some key conclusions for the study, and possible avenues for future work.

2. Quantifying the impact of wind energy on wholesale electricity markets

2.1. Economics of electricity supply and wind energy

A number of studies have presented the potential impact of wind generation in terms of its impact on the supply curve, or merit order, of generation within electricity spot markets (EWEA,

2009; Sensfus et al., 2007). Our framework presented here, provides a simplified representation of such markets for ease of interpretation, simplifying out potentially relevant factors such as transmission constraints, strategic bidding by generators, and potential demand-side participation, with the purpose of developing a set of hypotheses for the impact of wind on wholesale spot markets.

Fig. 1 presents stylised supply and demand curves for an electricity spot market. The demand curve, which represents the demand of electricity retailers and often large direct consumers, is typically assumed to be very steep or even vertical (inelastic), due to the fact that the demand for electricity by end consumers is often near independent of the spot price in the short term. The supply curve represents the aggregated offers of each generator to provide particular quantities of power (MW) if the market price is at or above a specified amount. In theory, assuming that no market power is present, the supply curve is equivalent to the aggregate marginal cost curve of all generators. However, this is almost never the case, especially at high levels of demand when there is tight supply and generators with non-dispatched capacity have freedom to charge higher prices due to lack of competition (Twomey, 2010). This results in the dramatic slope of the supply curve at high levels of demand. Depending on the particular market, the combination

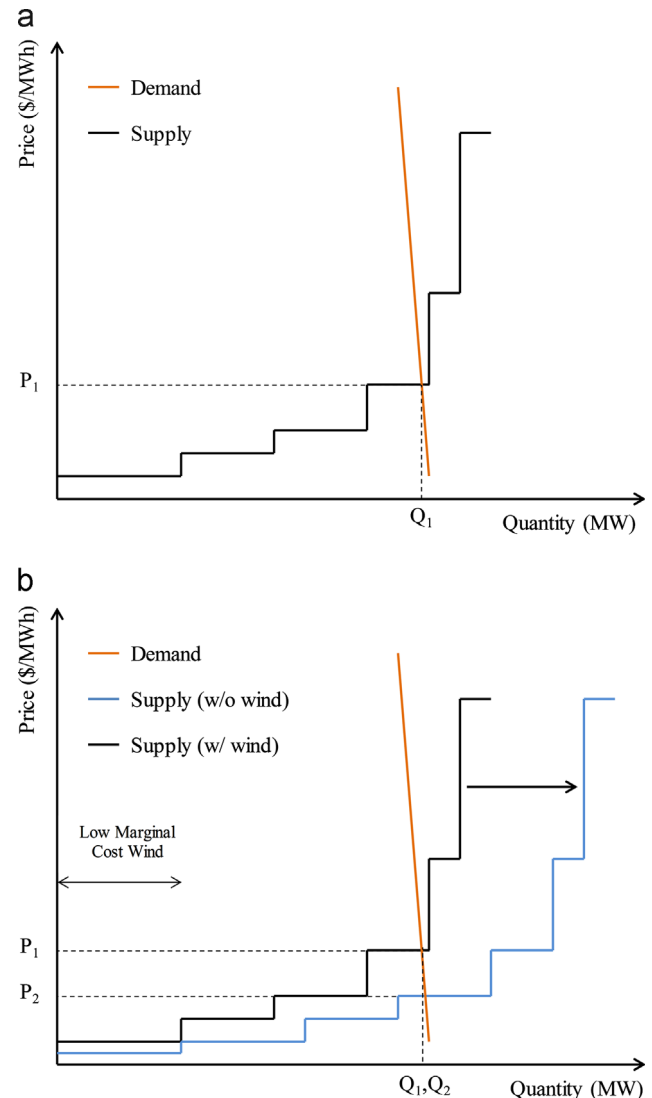


Fig. 1. Stylised supply and demand curves for spot market with inelastic demand curve. (a) without wind output and (b) with wind output.

of a near inelastic demand curve and steep supply curve results in the characteristic extreme spot price volatility.¹ Other complexities include the minimum operating levels, maximum ramp rates and significant unit commitment costs and time lags of many large thermal units which will often see them offer some of their output into the market at negative prices. In general, there is significant clustering of plants in the merit order depending on their primary fuel and technology—for example, from typically ‘free’ hydro and wind through to nuclear, coal, gas and expensive oil. The actual merit order is very market specific depending on the mix of generation and nature of fuel provision.

2.1.1. Potential wind impacts on spot market price

Fig. 1a and b shows how, all else being equal, wind tends to shift the supply curve due to its low marginal cost of generation. As Fig. 1b depicts, the shift in the supply curve due to wind causes the price to move from P_1 to P_2 where $P_2 < P_1$. Further, due to the increasingly steep shape of supply curve the magnitude of the difference between P_1 and P_2 will increase as demand increases. Finally, the variance of price is expected to increase due to these price changes caused by fluctuations in wind (Woo et al., 2011; Jacobsen and Zvingilaite, 2010).

It is important to note that depending on the treatment of wind energy in the dispatch process of a market it may be more accurate to interpret the impact of wind as a negative shift in the demand curve. That is, if wind is included in the centralised dispatch process (semi-scheduled in the NEM) it should be considered as supply side and if the wind energy is not included and given special generation rights (non-scheduled in the NEM) it should be considered to be a negative demand impact. The effect of these two treatments will be the same assuming that wind is always dispatched under both scenarios.²

These predictions, as noted in other studies (Woo et al., 2011; EWEA, 2009), lead to a number of hypotheses for the analysis of the wind impact on spot price. First, that price will be negatively correlated with wind output, second, that the magnitude of the impact of wind on price will increase with increasing levels of demand, and finally, that price volatility will increase with increasing levels of wind capacity due to the high variability of wind generation.

2.1.2. Potential wind impacts on other generator dispatch

Fig. 2a and b presents simplified dispatch scenarios in a market with purely wind, gas and coal fired generation. In these scenarios, wind generation has a strictly lower marginal cost of generation than coal fired generation which, in turn, has a strictly lower marginal cost than gas fired generation. As can be seen with the introduction of wind output, Q_{Wind} , in Fig. 2b, the quantity of gas required to dispatch, Q_{Gas} , is reduced, relative to Fig. 2a, due to its position in the merit order. In addition, one can imagine a scenario where demand is at a sufficiently low level and wind output sufficiently high such that no gas generation is required to dispatch ($Q_{Gas} = 0$) and therefore coal fired generation starts to be displaced. This phenomenon is one that is currently being observed in the South Australian (SA) and Victorian (VIC) regions of the NEM and is contributing to the occurrence of

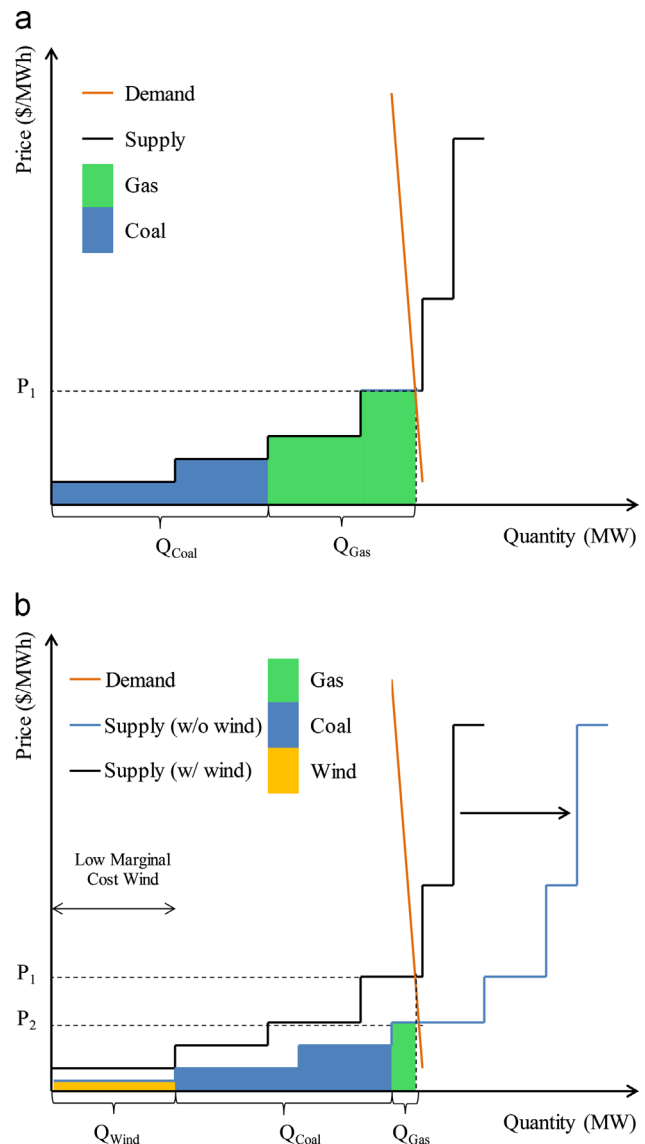


Fig. 2. Supply curve showing simplified generation merit order for SA. (a) without wind and (b) with wind.

negative prices due to baseload generators placing negative bids for outputs below their minimum operating level. From this framework it is hypothesised that higher cost peaking and intermediate generation will be displaced first, and lower-cost, baseload generation will be displaced at times when the majority of higher cost generation has been displaced from the market or is at minimum operating levels.

2.2. Studies to date estimating the price and dispatch impacts of high wind penetrations

There are a number of approaches to understanding interactions within electricity markets, each with different strengths, weaknesses and applicability. Broadly speaking, they range from theoretical modelling or simulation, which aim to simplify market interactions so that behaviour under different conditions can be simulated, to direct analysis of market data or empirical analysis, which aims to investigate actual participant behaviour through statistical or econometric analysis. Each method must apply a range of simplifying assumptions to enable clear conclusions and policy recommendation and all have limitations given the challenges, complexities and uncertainties in electricity industry operation.

¹ For example, on 31 January and 1 February 2011, Victoria experienced prices in excess of \$5000/MWh in three 5-min trading intervals. During this time there was no dispatch offer between the \$300/MWh and \$9300 MWh and so prices were extremely sensitive to changes in import capacity, capacity and demand (AER, 2011a).

² In the NEM, this assumption can breakdown during periods of low demand and high wind generation, where wind farms may choose to reduce output given negative prices resulting from the offer strategies of large thermal plant seeking to stay on-line, or periods where network constraints are binding (Cutler et al., 2011).

As one would expect, analyses can be very context specific. A key issue is the mix of existing plant within the power system – for example, wind generation can generally be expected to pose less challenges for systems with high levels of flexible, energy constrained, hydro generation by comparison with systems that have predominantly large and relatively inflexible coal or nuclear plant, such as the NEM. Also important is the level of interconnection with other grids and the geographical dispersion of the wind generation across the power system.

To date many of the studies looking at these impacts have used simulated models of an electricity market rather than attempting to estimate these effects through empirical analysis of market data. For example, [Sensfus et al. \(2007\)](#) utilised the calibrated model of German market to analyse the merit order effect in the market. Simulating the market with and without the Renewable Energy Sources Act (EEG) tariffs program for the years 2001 and 2004–2006, the estimated effect was found to range from \$2.26/MWh in 2001³ to \$10.44/MWh in 2006. [Delarue et al. \(2009\)](#) utilised a unit commitment simulation model to estimate the impact of wind generation on fuel cost reductions and reductions in carbon emissions in the Belgian electricity market, concluding that the total fuel saving from wind generation was approximately \$30.59/MWh. Furthermore, under conditions with high wind and low demand, simulations suggested baseload generation will be displaced while in other demand and wind combinations gas generation is mostly displaced. [Weigt \(2008\)](#) conducted an analysis of the German market for the potential impact of wind output on fossil fuel generation utilising a market equilibrium model to estimate the differences in production costs and market prices due to wind. The study concluded that on average wind caused a \$13/MWh drop in market prices. [Green and Vasilakos \(2010b\)](#) studied the impact of wind energy on optimal generation mixes in the British electricity industry. Modelling concluded that in the long run the impact of wind on price will be largely negated by the increase in more expensive flexible generation, which suggests that while wind can be considered to be reducing price in the short term due to the merit order effect, over time the reactions of other firms will tend to counteract these reductions.

Econometric techniques have been used extensively to model electricity prices but only in the recent times have they been utilised to investigate the effects of wind generation on electricity prices. [Woo et al. \(2011\)](#) used regression modelling techniques to look at the impact of wind generation on balancing market prices and price variance in the Electric Reliability Council of Texas (ERCOT) market in Texas, USA. An autoregressive model, lagged $\log(\text{price})$, was used to investigate the impact of wind output on price while controlling for other exogenous factors, such as electricity demand, gas price, non-dispatchable nuclear generation and seasonal factors. Results showed that a 1 MWh increase in wind led to a drop in spot price between 1.30c/MWh and 4.40c/MWh, the higher value being in the region with the highest wind penetration. In a similar analysis of balancing market prices in Texas, [Nicholson et al. \(2010\)](#) investigated the relationship between wind output and price in a series of autoregressive moving average models with exogenous variables (ARMAX) for seasonally adjusted price. The drop in price as a result of wind output was estimated to be between 0.35c/MWh and 1.64c/MWh per MWh of wind output.

2.2.1. Summary

There are limited studies employing econometric analysis techniques to estimating the impact of wind energy on wholesale

electricity markets. The analysis of the NEM presented in the next section is the first to look at these impacts in a market featuring a regional gross pool market and with consideration of wind generation on other types of generation as well as prices. From the above theoretical framework and survey of the literature we will test the following hypotheses for the effect of wind output on price and generator output in the two regions of the NEM with the most significant wind penetration, SA and VIC:

1. wind generation is negatively correlated with price;
2. wind generation will have a larger negative impact on price at higher levels of demand;
3. wind generation will have a significant negative correlation with peaking plant output once fluctuations in demand are accounted for; and
4. wind generation will have a marginal negative impact on baseload plant output, most noticeable during periods with low demand and high wind output.

This study will not look in depth at the volatility of spot prices induced by the presence of wind. Due to the extreme volatility of NEM spot prices, adequate modelling of volatility would require advanced techniques and would distract from the salient points of this paper. Also, as this study will focus on a truncated data set to provide insight into the normal operation of the market, any analysis of volatility would provide limited insight given that the volatility has been reduced through the truncation. We discuss the other necessary assumptions and limitations of this analysis approach in [Section 4](#).

3. The National Electricity Market

The Australian National Electricity Market (NEM) includes the states of New South Wales, Queensland, Victoria, South Australia and Tasmania and represents over 90% of Australian electricity demand. The NEM has many unique physical characteristics including its large, and largely north–south, geographic coverage (around 5000 km from Port Lincoln in South Australia to Port Douglas in Queensland) and the prominence of coal fired generation, including significant very low operating cost brown coal (lignite) plant. In Australia, electricity generation produces approximately 37% of national greenhouse gas emissions ([DoCC, 2009](#)), in large part because more than 80% of Australia's electricity generation comes from coal fired power plants ([AER, 2010](#)).

The wholesale spot market is unique in that it is 'energy only', with no capacity market or technical forward market, is compulsory (gross pool) for all generators larger than 30 MW, uses 5 min dispatch with 30 min averaged pricing, requires participants to manage their own unit commitment and allows participants to rebid their price-quantity offers on a 5 min basis. The spot market is characterised by its extremely volatile prices by comparison with many other markets and during the study period operated under a market ceiling price in any half hour of \$12500/MWh and a market floor price of –\$1000/MWh. At present, each of the regions in the NEM has only limited interconnection with other regions and therefore each region has its own dynamic wholesale prices ([AER, 2010](#)).

The wind energy industry in Australia has expanded significantly since the introduction on the Mandatory Renewable Energy Target (MRET) in 2001, as [Fig. 3](#) depicts. The scheme included the introduction of the Renewable Energy Certificate trading scheme, the first national scheme of its kind in the world ([MacGill et al., 2006](#)). Renewable Energy Certificates provide additional income for renewable energy generators on top of

³ This paper assumes exchange rates of 1.33 AUD/EUR and 1 AUD/USD. All dollar values are in USD unless otherwise specified.

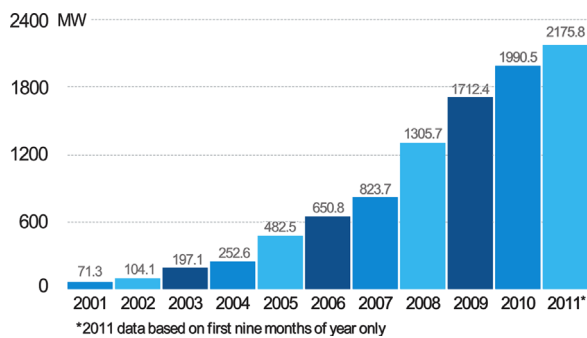


Fig. 3. Installed wind capacity in the NEM (Clean Energy Council, 2011).

the income from the wholesale spot market (and any derivative market activities). In July 2009, the Australian Government announced the Expanded Renewable Energy Target with a goal to reach 20% of Australia's electricity supply from renewable sources (Office of the Renewable Energy Regulator, 2011).

In the NEM, the largest proportions of wind generation are in SA and VIC with 48% and 23% of national installed wind capacity respectively. SA now has approximately 1200 MW of wind capacity in a region with average demand of approximately 1650 MW, while Victoria has approximately 400 MW of wind capacity with average demand of approximately 5950 MW.⁴ These represent wind penetrations (in terms of actual wind generation as a proportion of regional demand) of around 17% and 2% respectively. These levels of wind are impacting the operation of the market as has been noted by the market regulator, operator and a number of key studies (AER, 2011b; AEMO, 2007; Cutler et al., 2011; MacGill, 2010). In recognising the increasing influence of wind on the dispatch of other generators, the Australian Energy Market Operator (AEMO) created a new classification for generators, 'semi-scheduled', which incorporates wind energy into the centralised dispatch process where previously they operated unrestrained with 'non-scheduled' classification (AEMO, 2007).⁵

4. Model

4.1. Assumptions and variable selection

Some key statistical and economic assumptions are required for this analysis. The first assumption is that explanatory variables are exogenously determined or determined outside the market. This means that a one way causal relationship can be established between a variable and the spot price or level of generator output and that the model therefore does not suffer from endogeneity bias. In the case of wind output this is a generally valid assumption as wind generation is driven by the natural phenomenon and its low marginal cost means that it will be dispatched under the vast majority of market conditions. In the case of demand, exogeneity requires the economic assumption that demand is perfectly inelastic, meaning consumers of electricity in the short run do not change their consumption behaviour in response to price changes. This is considered to be a reasonable assumption since the majority of end consumers do not observe the spot market prices since retailers are the ones formally participating in the market.

The next assumption is that variables omitted from the analysis are largely uncorrelated with the explanatory variables used. Any such correlations are assumed negligible in this analysis and therefore it is assumed that no omitted variable bias is present. Other possible explanatory variables include gas spot prices and ambient temperature. Gas spot prices have not been included due to data availability but their omission is expected to only have a modest effect as firstly, gas supply is generally provided through long term supply contracts and secondly, from a modelling perspective their correlation with other explanatory variables is likely to be very low. Temperature was not included due to possible multicollinearity issues due to a high correlation with demand.

Demand values for surrounding regions have been included in each model in an attempt to capture the interregional effects caused by interconnection. While their inclusion increases the accuracy of the estimates of the effect of wind output, the inclusion of these variables means that interpretation of the coefficients of each of the individual demand variables is problematic due to multicollinearity issues caused by their high correlations. As the variable of concern is the coefficient of wind output this was deemed to be a worthwhile sacrifice. Wind output has been included in a single variable equal to the sum of output across the two regions. This arrangement avoids interpretation issues as a result of multicollinearity caused by the correlation between the wind output across the two regions.⁶

Therefore the explanatory variables that have been included in the analysis are: (1) the half-hourly total wind output (the sum of scheduled and non-scheduled across both SA and VIC); and (2) the half-hourly 'native' demand⁷ in SA and VIC and the neighbouring New South Wales region (Table 1).⁸

For each of the data series used in the analysis the Phillips-Perron test for stationarity was conducted and all series were found to be stationary as expected, thus alleviating the concern of spurious regression results.

4.2. Price model

The characteristics of electricity price data, and in particular NEM price data, provide a unique set of modelling issues that need to be understood and dealt with in order to undertake our analysis. In electricity price modelling the extreme volatility caused by price events and the prevalence of negative prices means that truncation of the data, or spike pre-processing, is a common procedure used when performing statistical modelling (Weron, 2006) and one used in price modelling and analysis in the NEM previously (Cutler et al., 2011; Thomas and Mitchell, 2005). The justification is that these events are considered 'outliers' and not indicative of the prevailing trend in the data. In particular, during extremely high and low prices, the market can be considered to be operating in a fundamentally different manner as prices deviate significantly from bidding according to short run marginal costs. At high prices, prices can be an indicator of a shortage of supply while at low or negative prices, prices are generally a result of demand approaching the minimum generation levels of baseload plants and the bidding of wind generators which are subsidised by renewable energy certificate revenue. In addition, they are often caused by rare events such as

⁶ The small amounts of wind generation in surrounding regions have been assumed to have a negligible effect on SA and VIC prices.

⁷ 'native' demand is defined as demand seen by the market dispatch process plus non-scheduled generation since non-scheduled generation is treated as negative demand and is subtracted off before entering the dispatch process.

⁸ Demand for Tasmania was not included as due to its small size the variable was not statistically significant.

⁴ Figures based on market data as of 3/2011.

⁵ Prior to the introduction of the 'semi-scheduled' classification some SA wind farms actually participated as scheduled plants the same as any other conventional generator (MacGill, 2010).

Table 1
Demand and output variable characteristics.

Variable	Max	Min	Mean	Standard deviation
30-min SA demand (MW)	3409.13	978.37	1658.90	315.81
30-min VIC demand (MW)	10 072.90	3888.37	5952.84	884.99
30-min NSW demand (MW)	14 745.86	5691.67	8868.83	1399.74
30-min SA wind output (MW)	972.44	−1.20	284.40	204.38
30-min VIC wind output (MW)	353.95	−2.45	113.14	90.45
30-min SA gas output (MW)	2494.33	158.00	777.75	289.76
30-min VIC gas output (MW)	1753.49	0.00	127.05	234.78
30-min SA coal output (MW)	736.33	0.00	526.94	127.98
30-min VIC coal output (MW)	6756.78	4359.49	5886.14	391.56

Table 2
Price variable characteristics.

Region	Variable	Mean	Min	1st-percentile	Median	99th-percentile	Max	Standard deviation	% Market revenue
SA	<i>price</i>	\$44.71	−\$996.7	\$4.14	\$25.10	\$93.66	\$12199.53	\$394.96	100
	<i>price</i>	\$28.57	\$1	\$4.14	\$25.10	\$93.66	\$415	\$26.66	50.9
	<i>log(price)</i>	3.204	0	1.42	3.223	4.54	6.03	0.534	50.9
VIC	<i>price</i>	\$32.15	−\$817.03	\$9.31	\$23.50	\$84.16	\$9998.60	\$200.24	100
	<i>price</i>	\$26.74	\$1	\$1	\$23.50	\$415	\$415	\$20.86	78.3
	<i>log(price)</i>	3.177	0	0	3.157	6.03	6.03	0.427	78.3

Note: || denotes truncated.

transmission network or generator outages, making it reasonable to treat them differently to periods that are not affected by these events. The use of truncated price data therefore analyses the impact of wind under a set of 'typical' operating conditions where prices that approximate short run marginal costs are present. While this does not reflect the entire market (see Table 2), it will still provide valuable insight into the relationship between wind output and the spot market price.

The price data was truncated between the upper bound of \$415/MWh and lower bound of \$1/MWh, with values outside this range set to the bounding values. The value of \$415/MWh was chosen as it is the estimated highest marginal cost of generation in the NEM and \$1/MWh as it approximately represents the lowest marginal cost in the NEM (ACIL Tasman, 2009). Electricity prices also possess seasonal trends at daily, weekly and yearly time scales and so in this model, seasonal components have been modelled using a series of binary values, one for each hour of the day in each season, representing 96 variables, as well as a binary indicator for weekends. The use of different daily binary variables for each season reflects the different daily profiles that occur each season.

An autoregressive term was included in the model to reduce the effects of any autocorrelation in the price data and to incorporate the effects of transient changes in market conditions.

Therefore, the price model is written as

$$\ln(\text{price}_t) = c + \gamma \ln(\text{price}_{t-1}) + \alpha_1 \text{wind}_t + \sum_{i=1}^3 \beta_i \text{demand}_i + \sum_j \mu_j S_{j,t} + \eta_1 W_t + \varepsilon_t \quad (1)$$

where $\ln(\text{price}_t)$ is the logarithm of truncated regional price, c is the constant term, γ is the coefficient of the lagged price term, α_1 and $\{\beta_i\}$ are the coefficients of the explanatory variables, i denotes the regions (1=SA, 2=VIC, 3=NSW), $\{\mu_j\}$ is the series of coefficients of the seasonal binaries, $\{S_{j,t}\}$ is the series of seasonal

binaries, η_1 is the coefficient of the weekend binary variable, W_t is the weekend binary variable and ε_t is the error term.

4.3. Generation models

4.3.1. Gas model

Given its flexibility and high marginal costs, natural gas output levels can generally be expected to be significantly affected by wind output. In terms of the supply and demand framework due to the fact that gas generally has a higher marginal cost than coal, it will be expected to be displaced first as the supply curve shifts due to wind output.

The model for gas output is written as

$$\text{gas}_t = c + \alpha_1 \text{wind}_t + \sum_{i=1}^3 \beta_i \text{demand}_i + \varepsilon_t \quad (2)$$

where gas_t is gas output for the region, c is the constant term, α_1 and $\{\beta_i\}$ are the coefficients of the explanatory variables, i denotes the regions (1=SA, 2=VIC, 3=NSW) and ε_t is the error term.

4.3.2. Coal model

Brown coal fired power plants, as found in SA and VIC, mostly have operating costs that are significantly lower than the gas fired generation also found in the regions. They also typically have relatively high minimum operating levels and significant startup and shut down costs. Therefore, it is expected that they would be the last type of plant to be displaced by wind generation. As the most greenhouse emissions intensive form of generation currently in large scale use around the world, a reduction in brown coal output represents a potentially significant outcome for the positive impact of renewable energy generation.

As discussed in Section 2.1 an important element of the relationship between wind and coal fired generation, or any baseload generation, is when high wind and low demand means that the demand to be met by conventional generation is so low as to require that baseload plant dispatch at levels lower than its

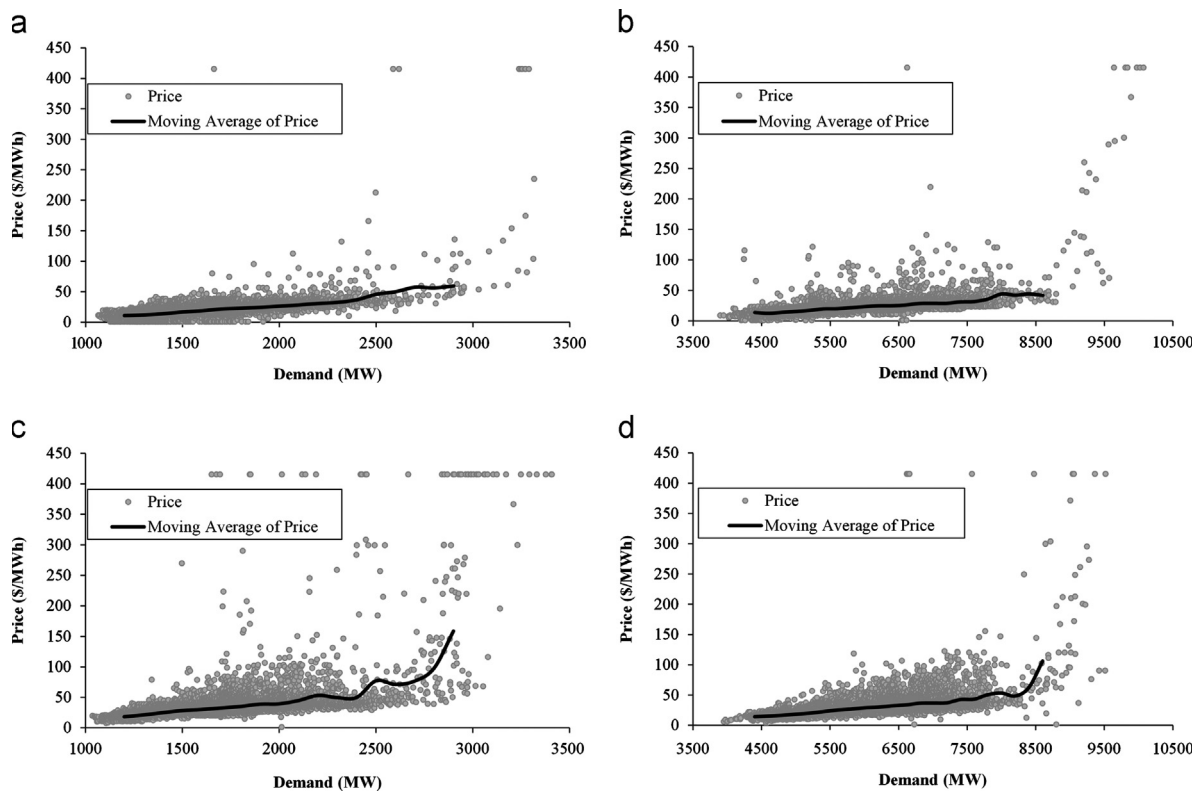


Fig. 4. Implied average supply curves for SA and VIC under the top and bottom quartiles of wind output. (a) SA, Top Quartile Wind, (b) VIC, Top Quartile Wind, (c) SA, Bottom Quartile Wind and (d) VIC, Bottom Quartile Wind.

usual output. It is expected that it will be under these circumstances that the majority of the displacement of baseload generation will occur. To incorporate this in the model of coal output a binary variable was added to the model to represent the expected effects on coal output under low demand and high wind scenarios. The model included the variable $HWLD_t$ which is 1 in periods when demand is in the bottom quartile and wind is in the top quartile and zero otherwise.

Therefore, the model is written as

$$coal_t = c + \alpha_1 wind_t + \sum_{i=1}^N \beta_i demand_i + HWLD_t + \varepsilon_t \quad (3)$$

where $coal_t$ is coal fired output for the region, $HWLD_t$ is the binary indicator for periods with top quartile wind and bottom quartile demand and the other symbols are as previously used.

5. Data

The data for this analysis is 30 min price, demand and output data for the SA and VIC regions of the NEM from 1/3/2009 to 28/2/2011. The data was extracted from the database located on the Australia Energy Market Operator (AEMO) website (www.aemo.com.au) with assistance from the NemSight analysis software.⁹

As Table 2 depicts, the truncation of the data means that a lower proportion of market revenue is incorporated into the analysis. As noted in Cutler et al. (2011) these high price events may be very uncommon but they can contribute to a significant proportion of total spot market revenue which represents a limitation to our price modelling approach.

6. Results – wind impact on price

6.1. Preliminary analysis

A preliminary understanding of the merit order effect and the relationship between price and wind output can be gained by plotting price versus demand for the top and bottom quartiles of wind for SA and VIC (Fig. 4). The moving average of price is superimposed on the plot to show the prevailing trend in the data and Fig. 5 shows a comparison of the moving averages under the high and low wind scenarios. In terms of the supply and demand framework, this shift in the moving averages can be interpreted as a shift of the supply curve to the right during times of wind output due to its low marginal cost, thereby effecting the price at which demand and supply are met. Clearly, due to the spread of points in the scatter plots shown there are other factors that drive change in the supply profile of the market. The aim of this analysis is therefore to quantify this price shift as a function of wind and demand.

6.2. Model estimation

In order to account for the fact that the data has been truncated a Tobit model has been used to estimate the parameter coefficients in the model. Table 3 presents the results of the model of spot price. The models of SA and VIC both show that wind output has a notable effect on price. This study estimates the merit-order effect to be a difference in price of approximately $-0.069\%/MWh$ ($-1.74c/MWh$ at the median price) for SA and $-0.027\%/MWh$ ($-0.62c/MWh$ at the median price) for VIC per MWh of total wind output across both regions.

These results confirm the first hypothesis of this study; that wind is negatively correlated with price. Further, it is broadly consistent with values obtained in other studies: Woo et al.

⁹ See www.analytics.com.au.

(2011) that estimated an effect between -1.30c/MWh and -4.40c/MWh per MWh of wind output and Nicholson et al. (2010) who estimated between -0.35c/MWh and -1.64c/MWh per MWh of wind output. Further, the fact that the logarithmic transform of the price data provides the best fit, suggests that wind has a higher effect on price at higher levels of demand thus providing evidence for the second hypothesis that the effect wind on price will be larger at higher levels of demand.

6.3. Model evaluation

To interpret the results of the model estimations and assess the model fit with actual prices the model was evaluated under scenarios assuming actual wind output and no wind output. Fig. 6a and b shows the distributions of the actual price data

and these predicted price values for both SA and VIC. As the graphs clearly show, the elimination of wind from the model shifts the distribution of prices upward in both SA and VIC. The significantly higher price shift found in SA is as expected due to both the higher penetration of wind in the region and the generally higher prices and marginal costs of generation.

6.3.1. Estimating the total merit order effect

From these results, it is possible to develop an estimate of the total merit order effect or the average price reduction as result of level of wind output in each of the regions by calculating the volume weighted average price difference over the analysis period. Table 4 shows the estimates produced for SA and VIC. As before, the significantly higher value found in SA is as expected due to both the higher penetration of wind in the region and the

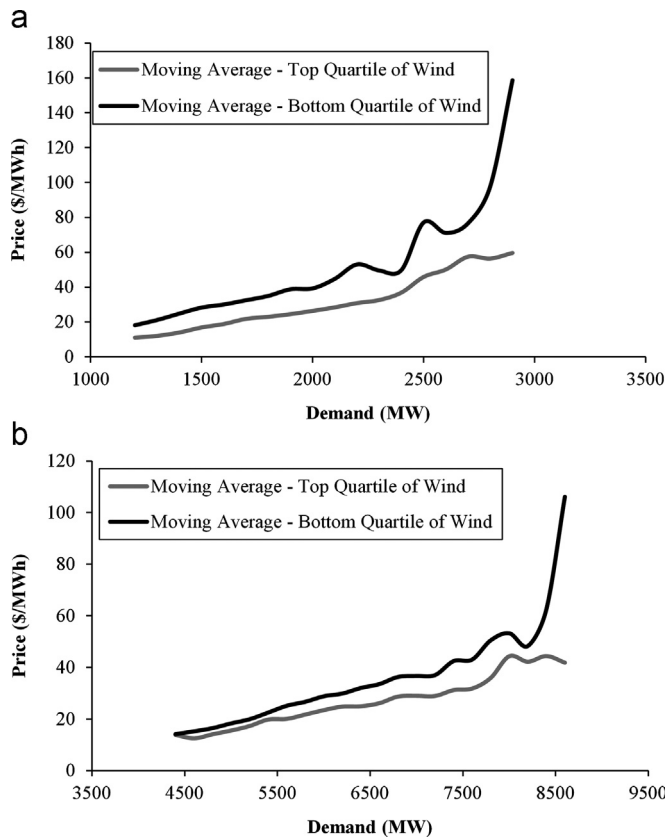


Fig. 5. Comparison of implied average supply curve under top and bottom wind quartiles. (a) SA and (b) VIC.

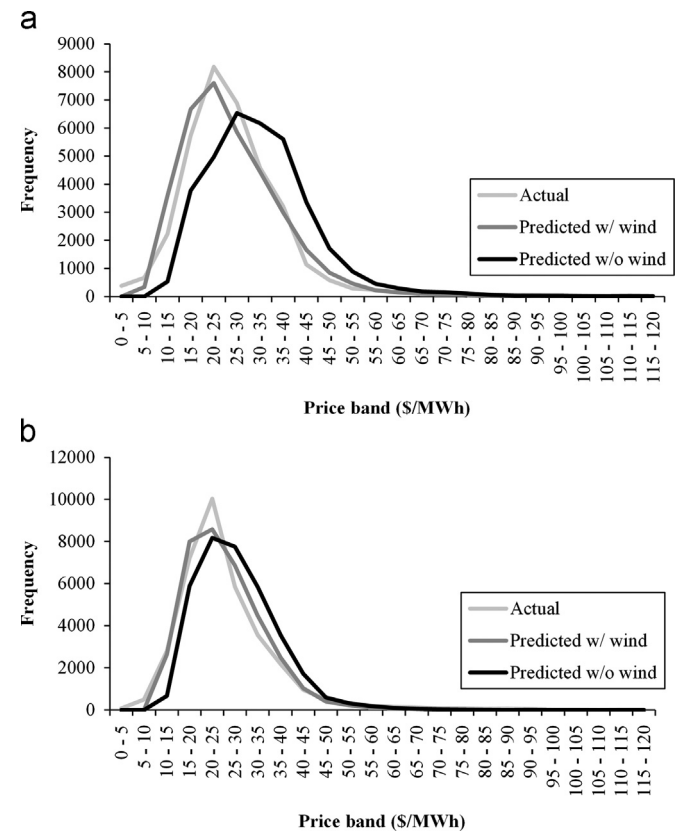


Fig. 6. Actual versus predicted price distributions with and without wind. (a) SA and (b) VIC.

Table 3
Price models – SA and VIC.

Region		SA		VIC	
Dependent variable		$\log(\text{price}_{it})$	SE	$\log(\text{price}_{it})$	SE
AR(1) term	γ	0.715 ^a	0.0034993	0.760 ^a	0.0020604
Constant	c	0.363 ^a	0.0201219	0.164 ^a	0.0146624
wind_{it}	α_1	-0.000197^a	0.00000515	-0.0000636^a	0.00000347
$\text{demand}_{t,SA}$	β_1	0.000138 ^a	0.00000096	-0.00000519	0.00000636
$\text{demand}_{t,VIC}$	β_2	0.0000675 ^a	0.00000391	0.0000716 ^a	0.00000258
$\text{demand}_{t,NSW}$	β_3	–		0.0000189 ^a	0.00000125
Merit order effect	$\frac{\alpha_1}{1-\gamma}$	$-0.069\%/ \text{MWh/MWh Wind}$		$-0.027\%/ \text{MWh/MWh Wind}$	

Note: Seasonal binaries have been omitted for simplicity.

^a Statistically significant at the 1% level (due to the possible presence of serial correlation, statistical significance was assessed by looking at the maximum of standard errors from the Tobit model and an OLS model with Newey–West standard errors).

Table 4

Estimates of the total merit order effect or the total change in average regional wholesale price per MWh due to wind generation.

SA	VIC
\$8.05/MWh	\$2.73/MWh

generally higher marginal costs of generation. A key assumption underpinning this calculation is that the impact of wind output on price is the sum of the short run impacts of wind on price. This calculation therefore does not take into account the long run changes caused by the impact of wind on price through entry or exit of other generators, changes in generation capacity or strategic bidding behaviour.

These figures are comparable with estimates of the total merit order of other studies, for example, *Sensfus et al. (2007)* that estimated between a value of \$2.26–\$10.44/MWh, and *Weigt (2008)* that estimated a value of \$13/MWh, both of which were conducted in the German electricity market.

6.4. Interregional effects

As a means of verifying the results obtained above and to gain a further understanding of the interregional impact of wind generation, further models were developed that modelled the SA and VIC regions and as a single joint region, that is, modelling the volume weighted price as a function of aggregate wind and demand. The two regions have significant transmission interconnection and share a somewhat similar wind regime *Cutler et al. (2011)* and therefore by aggregating across regions the flow between the regions, which can effect the estimated wind impacts for each region, can be ignored and we can gain a more holistic view of the total impact across the two regions. Through this analysis the total merit order effect was found to be \$4.28/MWh.

6.5. Further considerations

Recognising that truncation of price is not ideal due to the loss of information regarding extreme price events and that this may lead to biased estimates of the true merit order effect under all market conditions, other models were used that investigated the relationship between wind and raw price data. An Ordinary Least Squares regression was also run on the raw data but the fit of the model was extremely low and the coefficient of wind was counterintuitively not significant. In order to reduce the effect of outlying values a Least Absolute Deviation regression method¹⁰ was also used which produced a better R-squared value than the OLS estimation procedure but was still significantly lower relative to the truncated data case. Interestingly, using this estimation procedure the coefficient of wind was found to be statistically significant and the coefficient estimate was of the same order of magnitude as in the truncated data case but still the model had comparatively inferior explanatory power.

Analysis was also conducted on the impact of wind output on price spikes and negative prices. As expected, high wind output led to a higher frequency of negative prices and a lower frequency of high prices events. However, while wind output had a significant

impact on the frequency of these events, it had little observable impact on the magnitude. Further, the limited number of high and low price events means that the sample size is too small to make conclusive inferences around the relationship between wind and price event magnitude.

7. Results – wind impact on generation

7.1. Preliminary analysis

To gain a preliminary understanding of the relationship between wind and gas and coal fired output *Fig. 7* shows the average gas output across different quartiles of wind and demand. Generators that use natural gas as fuel in SA consist of both peaking gas turbines and intermediate gas fired boiler-type generators, hence the persistent level of gas fired generation at all times. In Victoria gas generation makes up a very small portion of the total installed generation and primarily consists of peaking gas turbines, hence the demand sensitive levels of output. *Fig. 8* shows the average coal fired output across different quartiles of wind and demand.¹¹ Both regions show the distinctive pattern of a small but noticeable drop in output under top quartile wind and bottom quartile demand, suggesting that under this circumstance wind is impacting the output of these generators.

7.2. Model estimation – wind impact on gas fired generation

Table 5 shows the results of the model estimation for the relationship between wind and gas for the two regions.

The total displaced generation was estimated by evaluating the model under scenarios of actual wind and demand and zero wind and actual demand. *Table 6* shows the results of the estimation of the reduction in gas generation attributable to wind output over the period in question.

Fig. 9 shows the actual demand profile and the predicted daily profiles under actual and no wind scenarios. As can be seen the predicted profile for gas output quite accurately describes the actual profile. The most notable deviation is in the VIC model where low demand overnight means that gas output is close to zero.

The analysis of the effects of wind output on gas generation shows that wind has a notable effect in both SA and VIC. The estimated total amount of displaced gas generation as a percent reduction is –12.1% in SA and –24.3% in VIC, leading to an estimated total reduction in carbon emissions of approximately 1.55 million tCO₂e over the study period. The higher reduction in gas output in VIC is expected as, even though SA has a higher wind penetration, gas fired generators in VIC are predominantly peaking whereas SA gas generators are both peaking and intermediate.

This analysis broadly confirms the predictions of simulation studies (*Green and Vasilakos, 2010a; Delarue et al., 2009*) that suggest there is significant potential for the displacement of peaking and intermediate generation. These results have also confirmed the third hypothesis that wind will have a significant effect on generators that are positioned at the upper end of the supply curve which in the context of the NEM is gas fired generation.

¹⁰ Least Absolute Deviation estimation method minimises the absolute value of the regression residuals ($\min \sum_{i=1}^N |e_i|$) rather than the squared residuals ($\min \sum_{i=1}^N e_i^2$).

¹¹ Note that the scale is cropped in *Fig. 8b*. The counterintuitive observed decrease in output between 2nd and 3rd quartile wind under top quartile demand, while small, is expected to be the result of a small localised correlation between wind and demand.

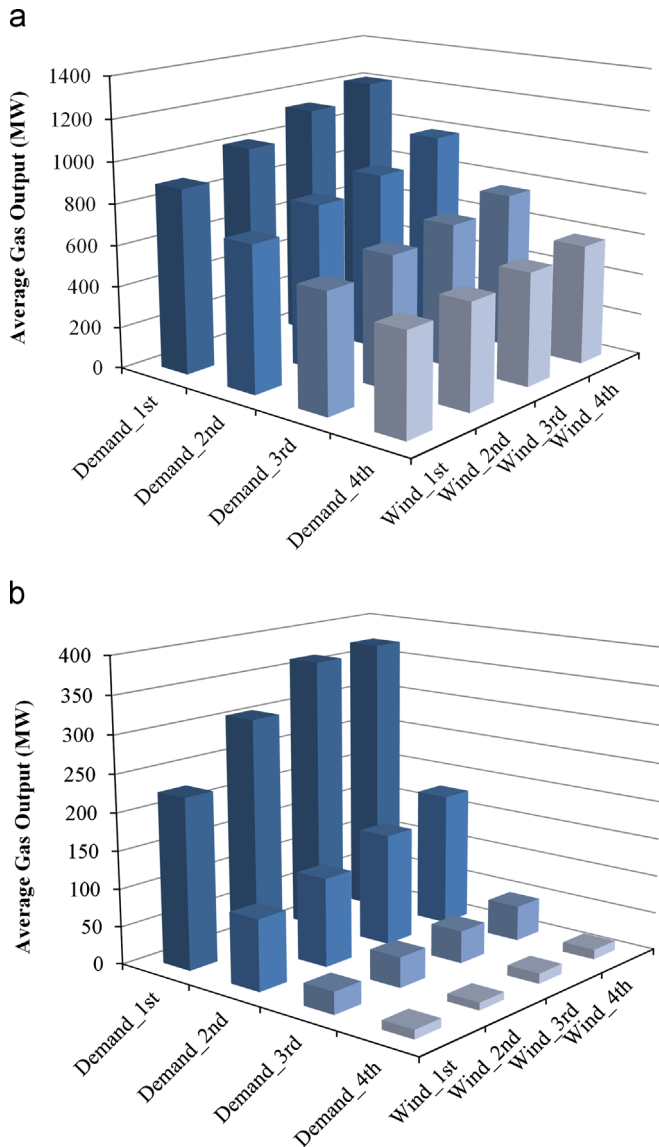


Fig. 7. Average natural gas output across wind and demand quartiles. (a) SA and (b) VIC.

7.3. Model estimation – wind impact on brown coal generation

The results of the model estimation for the wind impacts on coal fired generation are shown in Table 7. Table 8 shows the results of the estimation of the displaced coal generation attributable to wind output over the period in question.

Fig. 10 shows the actual daily demand profile and the predicted profiles under actual wind and no wind scenarios. The predicted daily output profile provides a relatively good approximation to the actual profile though the estimation of the dip in coal output overnight is underestimated. As is expected the model estimates that the highest reduction in coal output due to wind occurs overnight, when demand is at low levels.

The results of the analysis of wind impacts on coal generation provide strong evidence that wind output is displacing coal fired generation in both the SA and VIC regions of the NEM. While the estimated impacts are relatively small, with reductions of -2.03% in SA and -0.40% in VIC due to wind, they lead to a reduction in carbon emissions of approximately 0.8 million tCO₂e over the study period. On a per MWh of wind basis the impact is much larger in VIC than SA. This higher effect of wind on coal in VIC is expected due to the higher proportion of coal fired generation in

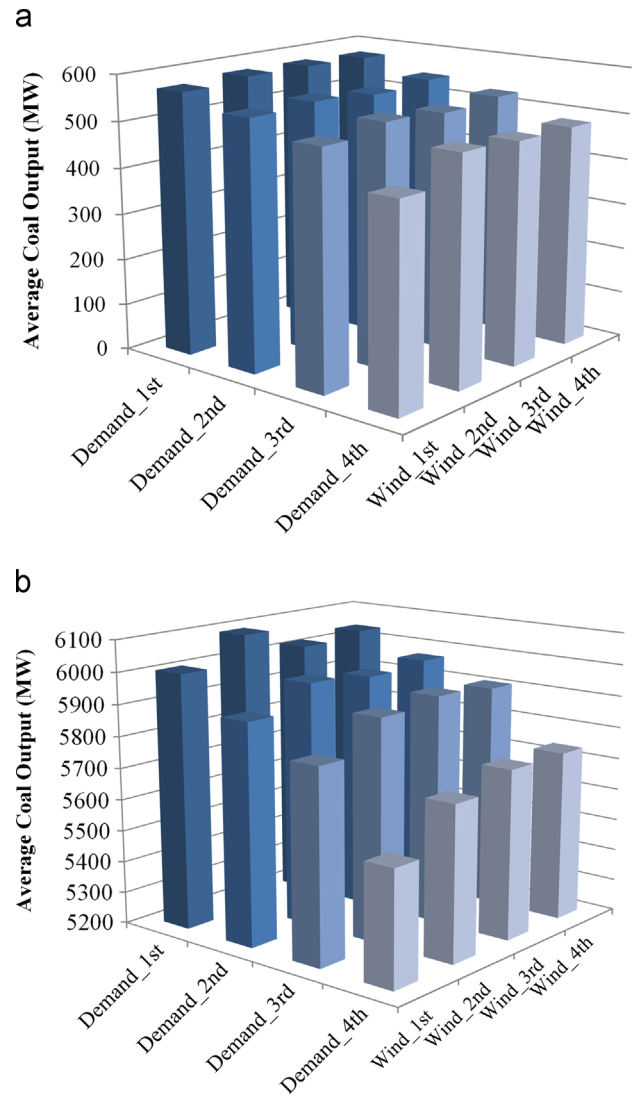


Fig. 8. Average brown coal output across wind and demand quartiles. (a) SA and (b) VIC.

Table 5

Gas model estimation results.

Region	Dependent variable	SA gas	SE	VIC gas	SE
R^2		0.7195		0.3819	
RMSE		153.46		184.59	
Constant	c	-502.32^a	8.047	-704.90^a	13.665
$wind_t$	α_1	-0.2702^a	0.0042	-0.1018^a	0.0051
$demand_{t,SA}$	β_1	0.4575^a	0.0082	0.2448^a	0.0129
$demand_{t,VIC}$	β_2	0.1056^a	0.0027	0.0453^a	0.0058
$demand_{t,NSW}$	β_3	–		0.0222^a	0.0034

Note: Newey–West standard errors used to eliminate any effects of serial correlation and heteroskedasticity.

^a Statistically significant at the 1% level.

VIC (90%). Analysis of cross-regional effects of wind output is problematic due to multicollinearity issues caused by high correlation of wind output in neighbouring regions, an issue discussed in the next section. These results confirm the simulation results of (Weigt, 2008; Delarue et al., 2009; Green and Vasilakos, 2010b) and our fourth hypothesis which suggested that wind output would have marginal effect on baseload generation.

Table 6
Gas model evaluations from 1/3/2009 to 28/2/2011.

Region	SA	VIC
Actual gas output (GWh)	13 626	2226
Predicted gas output at zero wind and actual demand (GWh)	15 496	2940
Predicted change in gas due to wind (GWh)	–1870	–714
Predicted change in gas due to wind (%)	–12.1%	–24.3%
Predicted change in CO ₂ e due to wind ^a (tCO ₂ e)	–1 122 037	–428 137
Predicted change in CO ₂ e per MWh wind ^a (tCO ₂ e/MWh wind)	0.16	0.06

^a Using 0.6 tCO₂e/MWh for carbon emissions of gas generation.

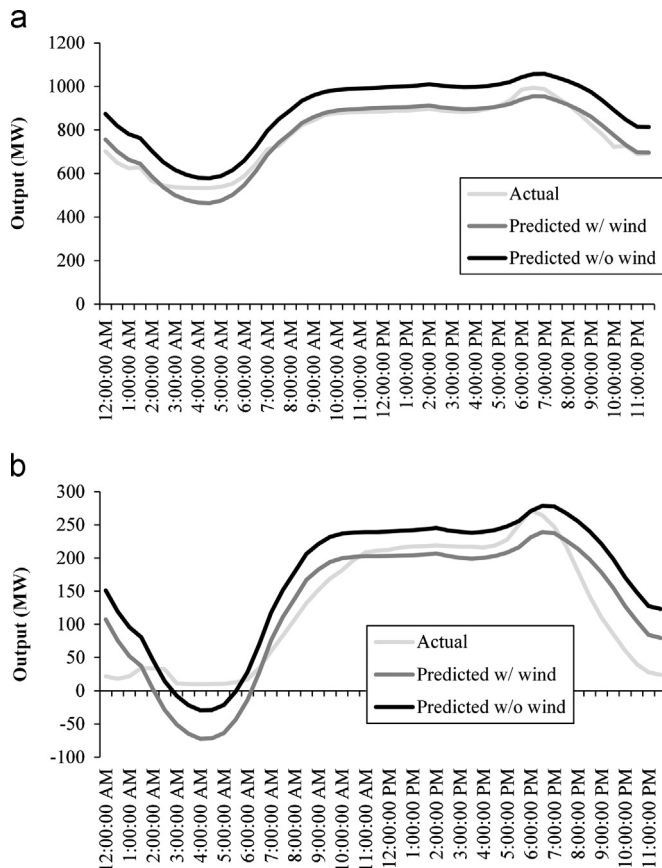


Fig. 9. Gas model actual versus predicted daily profile under actual and no wind scenarios. (a) SA and (b) VIC.

7.4. Interregional effects

In an attempt to eliminate the effects of the regional transfers on the estimated impacts wind in each region, similar to the analysis of price, a model for the joint SA and VIC regions was created. This model estimated the relationship between the aggregate wind and demand and the aggregate output from gas and coal fired generation. Table 9 shows a summary of the results obtained.

The estimated total reduction in output due to wind is lower compared to the case where the regions were modelled individually which is a result of the correlations between variables across the two regions. This means that modelling the regions separately while insightful in that it breaks down where the impacts are coming from it may produce overestimations and that in reality the true impact is in the range between the models in this section and Sections 7.2 and 7.3. Interestingly the total displaced generation across the two regions represents 44% of the

Table 7
Coal model estimation results.

Region	SA	SE	VIC	SE
Dependent variable	<i>coal</i>		<i>coal</i>	
R^2	0.0939		0.1364	
RMSE	121.83		363.91	
Constant	c	353.03 ^a	6.383	5107.96 ^a
$wind_t$	α_1	–0.0255 ^a	0.0036	–0.0499 ^a
$demand_{t,SA}$	β_1	0.0789 ^a	0.0066	0.0302
$demand_{t,VIC}$	β_2	0.00942 ^a	0.0021	0.0500 ^a
$demand_{t,NSW}$	β_3	–	0.0521 ^a	0.0040
$HWL_{t,t}$	η_1	–48.94 ^a	3.699	–165.65 ^a

Note: Newey–West standard errors were used to eliminate any effects of serial correlation and heteroskedasticity.

^a Statistically significant at the 1% level.

total wind output suggesting that 56% of wind generation is absorbed by other regions besides SA and VIC and other sources of fuel besides gas and coal, a statistic that reflects the role of interconnection right across the five regions of the NEM in facilitating wind integration.

8. Conclusion

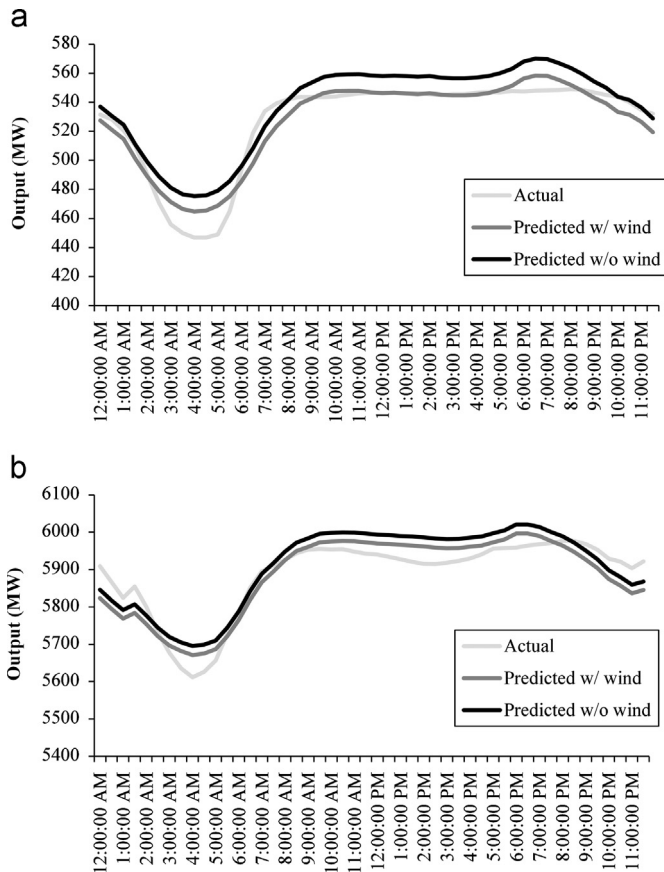
This paper has applied econometric analysis techniques to the estimation of the impact of wind output on prices and gas and coal fired generation. Based on the literature and a simple demand supply framework for electricity markets, we hypothesised that wind output will have a negative impact on price and one that is increasing with higher levels of demand and that such wind will displace primarily higher operating cost and more flexible gas generation and to a lesser extent coal fired generation. The analyses of data from the Australian NEM over the past two years supported these hypotheses. A natural extension of this result is that larger levels of wind installed capacity is likely to cause a more severe downward pressure on prices, decreasing the incentives for additional wind generation to enter the market. The implications of these results are numerous. As *Intelligent Energy Systems (2009)* notes, this economic limit may even become the limiting factor in the expansion of wind and, as *Cutler et al. (2011)* suggests, the South Australian region of the NEM may well be approaching this limit. In particular, these results suggest that price outcomes have been significantly altered due to the presence of subsidised wind generation and the methodology developed here provides a basis for assessing the costs and benefits of these renewable energy support schemes and analysing the financial implications for other generator types.

While in the short term there is an economic saving due to the lower marginal costs of wind generation, in the long term these savings are likely to be reduced. As *Green and Vasilakos (2010b)* concludes, the increased requirement for peaking generation to

Table 8

Coal model evaluation – 2 years from 1/3/2009 to 28/2/2011.

Region	SA	VIC
Actual coal output (GWh)	9232	103 125
Predicted coal output at zero wind and actual demand (GWh)	9423	103 538
Predicted change in coal output due to wind (GWh)	–191	–413
Predicted change in coal output due to wind (%)	–2.03%	–0.40%
Predicted change in CO ₂ e due to wind ^a (tCO ₂ e)	–267 825	–578 264
Predicted change in CO ₂ e due to wind ^a (tCO ₂ e/MWh wind)	0.038	0.083

^a Using 1.4 tCO₂e/MWh for carbon emissions of coal fired generation.**Fig. 10.** Coal model actual versus predicted daily profile under actual and no wind scenarios. (a) SA and (b) VIC.**Table 9**

Joint SA-VIC regional output model evaluation.

Dependent variable	Gas	Coal
Predicted change in output due to wind (GWh)	–2375	–420
Predicted change in output due to wind (%)	–13.0	–0.37
Predicted change in CO ₂ e due to wind (tCO ₂ e)	–1 425 199	–589 235
Predicted change in CO ₂ e due to wind (tCO ₂ e/MWh wind)	0.205	0.085

support the unreliability of wind will mean additional fixed costs that need to be absorbed into the market. The market will require more capacity and so prices may tend to move in an upward direction. For this reason, it is expected that the drop in prices will not persist in the long term though the differential in price received by wind and conventional plant is expected to remain. Other potential complexities include the possible impacts of high wind penetrations on derivative and ancillary service market

pricing, and the exercise of strategic market power. These are highly context specific issues and an important area for future work (MacGill, 2012) in order to assist market participants and policy makers in facilitating high penetrations of wind, and other highly variable and somewhat unpredictable renewable generation sources.

Additional further work could be to investigate the role of geographic diversification in the mitigation of the merit order effect on price. Such an analysis would be able to provide interesting insight into the value of a wind resource not only as a function of the expected capacity factor but also the correlation with existing wind resources. To supplement these direct analyses of market data more work in simulating the NEM and the impact of wind energy would be beneficial.

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